Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: What Statistics Is For

Every day you gather facts and figures around you — how many students were in class, how many bottles of water were sold, how many goals your favourite team scored. These facts and figures are called data. Data are facts or information collected for a particular purpose.

Statistics is the science of collecting, organising, presenting, analysing and interpreting data. When you count the pupils present in the morning and write their names in a register, you are collecting data. When you sort those names by class or by age, you are organising it. Statistics brings a clear order to information so that it becomes useful to you.

Why do we bother with statistics at all? Statistics helps us in five important ways:

- (a) Collect information — gather the facts you need.
- (b) Organise information — put the facts into a sensible order.
- (c) Compare information — see how one set of facts stands against another.
- (d) Make decisions — use the facts to choose sensibly.
- (e) Present information clearly — show the facts so others can understand them at a glance.

Consider a shopkeeper who sells oranges. Each day she counts how many oranges she sells (collecting information), and writes the daily totals in her notebook one after another (organising information). By looking at her notes she can compare Saturday's sales with Monday's (comparing information), and then she decides how many oranges to buy for next weekend (making decisions). When she wants to show her progress to her boss, she draws a simple table or chart so the figures are clear at a glance (presenting information clearly). Every one of these steps is statistics at work.

Practice

1. What is statistics?
2. What are data?
3. Name any three ways statistics helps us.

Section 2: Recording Data with Tally Marks and Frequency Tables

Statistics is built on data — the facts and numbers you gather when you count or record something. Before you can study data you have to collect it, and there are four common ways to do that:

1. **Observation** — you watch and note down what you see.
2. **Counting** — you count the items one by one.
3. **Interview** — you ask someone a question and record their answer.
4. **Questionnaire** — you give someone a written list of questions to answer.

Once you have collected your data, you need a neat way to record it as it comes in. That is where **tally marks** are used. A tally mark is a single stroke that stands for one item. You add one stroke for each item you collect, and every fifth stroke is drawn across the previous four. So five items are written as a bundle of four strokes with a fifth crossing them. Grouping the marks in fives makes counting fast, because you can count the bundles of five and then the leftovers.

Suppose you stand by the road and note the vehicles that pass, in this order:

bus, car, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car

You record each vehicle with a tally mark under its name as it passes. When you finish, your tally looks like this:

Vehicles	Tally	Total
car	||||/		6
bus			1
lorry	||	2
bicycle	|||	3
motor bike		0

Let's check the tally. **car** appears six times in the data, so it earns one full group of five (four strokes with the fifth across them) and one more mark on its own — that is 6, and the total column shows 6. **bus** appears once, **lorry** twice, and **bicycle** three times, so each gets the matching number of marks. **motor bike** never appears, so it gets no marks and its total is 0. Add the totals and you get $6 + 1 + 2 + 3 + 0 = 12$, which matches the 12 vehicles listed.

When your tally is complete, you can present it in a **frequency table**. A frequency table is a table showing how many times each item occurs in a set of data. It brings all the totals together into one clear summary.

Here is a frequency table showing the favourite colours of a group of students:

colour	frequency
Red	18
Blue	20
Green	14
Yellow	16
White	15
Pink	30

The **frequency** is simply how many students chose that colour. Reading this table, you can immediately see that Pink is the most popular colour (30 students chose it) and Green is the least popular (only 14 chose it). A frequency table lets you read the whole set of data at a glance, so you can compare items and answer questions about the data easily.

Practice

1. State three methods of collecting data.
2. Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.
3. A class recorded the favourite colours of its students in the frequency table below:

colour	frequency
Red	18
Blue	20
Green	14
Yellow	16
White	15
Pink	30

   (a) How many students chose Pink? (b) Which colour was chosen by the fewest students?

Check your work

Attempt every question before you read this page.

What Statistics Is For

1. Statistics is the science of collecting, organising, presenting, analysing and interpreting data.
2. Data are facts or information collected for a particular purpose.

3. Any three of: collect information, organise information, compare information, make decisions, present information clearly.

Recording Data with Tally Marks and Frequency Tables

1. Any three of: observation, counting, interview, questionnaire.
2.

number	tally	frequency
1	|||	3
2	|||	3
3	||||	4
4	|||	3

   Count how many times each number appears. The number 1 appears three times, 2 appears three times, 3 appears four times, and 4 appears three times. So the frequency table lists each distinct number with its tally marks and its total, and the totals add to $3 + 3 + 4 + 3 = 13$, which matches the number of data values.
3. (a) 30 students chose Pink. (b) Green, chosen by 14 students, was the fewest.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-f952a647	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____